arc_temp_humid - outstanding work

Updated 6 September 2026. The repository rename and the main-site move are complete and are no longer listed here; see `CHANGELOG.md` for what was done.

Everything described below is **not yet done**. What *is* done is in `CHANGELOG.md`; the architecture is in `CLAUDE.md`.

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1. Confirm the Holywell room type

The only thing still genuinely blocked. See section 2 - if `0E3C12EC` ("Internal Ambient", Holywell Barn) is a bedroom, TM59's 26 C night criterion applies to it; if it is a living space, no fixed threshold does.

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2. Decide the UK overheating threshold

Currently **none**: the three UK datasets have `"threshold": None`, so no red band is drawn and the sidebar control is hidden. Tanzania keeps 32-35 C.

Having now read CIBSE TM59 (2017), the position is clearer than "pick a number", and leaving it off may well be the right answer.

What TM59 actually requires

Compliance for a naturally ventilated home is **both** of (section 4.2):

- **(a) Living rooms, kitchens and bedrooms.** The hours where dT is at least 1 K, May to September, must be no more than 3% of occupied hours. This is TM52 Criterion 1, and dT is measured against the **adaptive** comfort limit, not a fixed temperature.
- **(b) Bedrooms only.** Operative temperature between 22:00 and 07:00 must not exceed **26 C** for more than 1% of annual hours. TM59 spells out the count: 32 hours, so 33 or more is a fail.

So the only fixed number in TM59 is 26 C, and it applies **to bedrooms, at night, counted over the year**. For a living room or kitchen there is no fixed threshold at all - the test is the adaptive one, which the adaptive comfort chart already serves.

What that means for the two UK sensors

Logger	Name	Room type	Fixed threshold applies?
`169502D1`	Living Room (No. 57), Grove	living room	**no** - criterion (a) only
`0E3C12EC`	Internal Ambient, Holywell	**unknown**	only if it is a bedroom

Grove's sensor is a living room, so TM59 gives it no fixed band. Holywell's is labelled only "Internal Ambient"; **that is the room type worth confirming**.

Three honest options

1. **Leave the threshold off** (current state). Defensible: neither sensor is a confirmed bedroom, and criterion (a) is already covered by the adaptive chart.
2. **Add 26 C only where a logger is a bedroom.** Correct, but the threshold is currently per-region, so this needs a per-logger threshold - a real change, though not a large one.
3. **Draw 26 C across the UK region as a rough reference.** Quickest, and the least honest: TM59's 26 C is night-only and bedroom-only, so a permanent band across a 24-hour living-room chart misrepresents the standard.

Two caveats before quoting TM59 numbers from this dashboard

- TM59 is a **design and modelling** methodology. It assumes standard occupancy profiles and dynamic simulation, and evaluates **operative** temperature. This dashboard plots **measured air** temperature, which is already listed as a known limitation in `adaptivecomfort.md`. Numbers taken from here are indicative of TM59, not a TM59 assessment.
- Criterion (a) is adaptive against **TM52 Category II**, whose upper limit is `0.33 x Trm + 18.8 + 3`. The dashboard currently offers ASHRAE 55 and the Vellei models, neither of which is that line. If the UK datasets are ever to be read against TM52/TM59, adding a TM52 Cat II band to the comfort model dropdown is the change that would do it - and is probably worth more than the threshold band.

3. UK Open-Meteo coordinates - DONE

Both feeds now use the **Open-Meteo model grid cell centre** for their building:

```
"grove":    {"lat": 52.057774, "lon": -2.704697, "start_date": "2026-07-01"}
"holywell": {"lat": 52.926643, "lon": -4.230377, "start_date": "2026-07-01"}
```

Requesting any point inside a grid cell returns that cell's series, so asking for the centre gives byte-identical data to asking for the building, while the number committed to this public repository is a fixed feature of the weather model rather than a private address. The cell centres sit 0.60 km (Grove) and 0.65 km (Holywell) from the buildings.

Worth recording: the town-centre coordinates used before were **already in the same grid cell** as both buildings, so the weather data was correct all along. The change is about what gets published, not about accuracy.

`start_date` is each building's first sensor reading (2026-07-31) minus a 30-day lead-in. The lead-in is not padding: the EN16798-1 running mean is seeded from its first day and decays that seed by alpha=0.8 per day, so without a run-up the first fortnight of comfort points would be measured against a running mean still carrying an arbitrary starting value. Thirty days reduces the seed's weight to about 0.1%, and in practice reproduces the running means obtained from three and a half years of history to within 0.1-0.3 C.

If sensors are ever installed earlier than July 2026, move `start_date` back to 30 days before the new earliest reading.

4. ARC UK Omnisense fetch - DONE, untested in CI

`SITES["uk"]` is filled in with site number **58345** ("Simmonds.Mills Retrofits") and a `--site uk` step runs in `update-dashboard-data.yml` alongside the Tanzanian one, using the same `OMNISENSE_USERNAME` / `OMNISENSE_PASSWORD` secrets.

It has not run yet. The credentials live in repository secrets and are not available locally, so the login and download path could not be exercised here. The first scheduled run (04:00 UTC) is the test. The step is `continue-on-error`, so a failure will not block the rest of the build; check the run log, and the sidebar's "Omnisense (UK) last updated" note, afterwards.

Until it succeeds the hand-added export in `data/omnisense_uk/` remains the source, and nothing breaks if the fetch fails.

5. Known limitation, not a task

ASHRAE 55 Section 5.4.1(a) requires that no heating is in operation. Nothing at any site records heating status, and nothing in a temperature and humidity trace reliably separates a heated room from a merely warm one.

The dashboard states this rather than inferring it: a bolded line in the sidebar applicability panel directly beneath the quoted criterion, and in the tooltips explaining the method. The 10-33.5 C running-mean gate is enforced automatically and greys out readings the standard does not cover.

That gate now has teeth it did not have before, because the UK Open-Meteo feeds reach back to March 2023: once UK winter indoor data exists, cold-weather readings will be greyed and excluded from the comfort percentage. But **a mild day with the heating on will still pass every test the dashboard can apply**.

Closing this properly needs a heating-status input, not a cleverer algorithm. See `adaptivecomfort.md` section 2 for the full reasoning, including why calendar-month filtering was considered and rejected.